Subject Code	:	CEUE203	Semester	:	3	Academic Year	:	2026-27
Subject Name	:	Object-Oriented Programming						

	Practical 1
Program: 1.1	**1. Vending machine** (VendingMachine.java, main in this file) (a) Define an enum Coin with constants ONE, TWO, FIVE, TEN. (b) In main, set a snack price of 15 and a running total of 0; create a Scanner. (c) Loop: read a coin name, use a switch expression to convert the Coin to its value (ONE→1, TWO→2, FIVE→5, TEN→10), add it to the total, and print the total so far. (d) Stop the loop once the total reaches 15 or more. (e) Print the change to return (total − 15). Expected output: TEN then FIVE prints "Paid. Change: 0"; TEN then TEN prints "Paid. Change: 5"

Code	`import java.util.Scanner;`
	`enum Coin`
	`{`
	`  ONE,TWO,FIVE,TEN`
	`};`
	`public class VendingMachine {`
	`  public static void main(String[] args)`
	`  {`
	`  int snack_price=15;`
	`  int total=0;`
	`  Scanner sc=new Scanner(System.in);`
	`  while(total<snack_price)`
	`  {`
	`  System.out.println("enter coin : ");`
	`  String coin=sc.next();`
	`  Coin coin_in = Coin.valueOf(coin);`
	`  int value=switch(coin_in)`
	`  {`
	`  case ONE -> 1;`
	`  case TWO -> 2;`
	`  case FIVE -> 5;`
	`  case TEN -> 10;`
	`  default -> 0;`
	`  };`
	`  total+=value;`
	`  System.out.println(" current total :" + total);`
	`  }`
	`  int rem=total-snack_price;`
	`  System.out.println("paid."+"change : "+rem);`
	`  sc.close();`
	`  }`
	`}`

Output

```
C:\materials\3rd sem\OOP java\practicals\practical 1>javac VendingMachine.java

C:\materials\3rd sem\OOP java\practicals\practical 1>java VendingMachine
enter coin : ONE
 current total :1
enter coin : FIVE
 current total :6
enter coin : TEN
 current total :16
paid. Change: 1
```

Program: 1.2

Toll booth (TollBooth.java, main in this file) (a) Define a record Vehicle(String number, String type). (b) In main, keep a running toll total and three counters (bike, car, truck). (c) Loop until the user types "done" for the number: build a Vehicle, use a switch expression on its type for the toll (bike→20, car→50, truck→150), add to the total, and increment that type's counter. (d) After the loop, print the total toll and the type with the highest count. Expected output: two cars + one bike → "Total toll: 120", "Most frequent: car".

Code:

```
import java.util.Scanner;
record Vehicle(String number,String type)
{

}
public class TollBooth {
   public static void main(String[] args)
   {
      int total=0;
      int car=0,bike=0,truck=0;
      Scanner sc =new Scanner(System.in);
      while(true)
      {
         System.out.print("Enter vehicle number : ");
         String num= sc.next();
         if(num.equals("done"))
         {
            break;
         }
         System.out.print("Enter vehicle type : ");
         String typ= sc.next();

         Vehicle v= new Vehicle(num,typ);
         int value=switch(v.type())
         {
            case "bike" ->
            {
               bike++;
               yield 20;
            }
            case "car" ->
            {
```

```
                car++;
                yield 50;
            }
            case "truck"->
            {
                truck++;
                yield 150;
            }
            default ->
            {
                System.out.println("invalid vehicle type");
                yield 0;
            }
        };
        total+=value;
    }
    System.out.println("total toll : "+ total);
    String most;
    if(car>=bike && car>=truck)
    {
        most="car";
    }
    else if(bike>=car && bike>=truck)
    {
        most="bike";
    }
    else
    {
        most="truck";
    }
    System.out.println("most frequently : "+ most);
    sc.close();
  }
}
```

Output:

```
C:\materials\3rd sem\OOP java\practicals\practical 1>javac TollBooth.java

C:\materials\3rd sem\OOP java\practicals\practical 1>java TollBooth
Enter vehicle number (done if complete) : GJ11AB1234
Enter vehicle type : car
Enter vehicle number (done if complete) : GJ18AR6542
Enter vehicle type : bike
Enter vehicle number (done if complete) : GJ20CV654
Enter vehicle type : truck
Enter vehicle number (done if complete) : GJ13DV6549
Enter vehicle type : car
Enter vehicle number (done if complete) : done
total toll : 270
most frequently : car
```